SJSU SAN JOSÉ STATE UNIVERSITY

Charles W Davidson College of Engineering · Computer Engineering

Data Structures and Algorithms

CMPE 180A

Spring 2026 Section 01 In Person 3 Unit(s) 01/22/2026 to 05/11/2026 Modified 01/21/2026

Contact Information

Dr. Juan Gomez

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Office: ENG 281

Course Information

Office hours: We 5:00-6:00pm

Section:

Course materials such as syllabus, handouts, notes, assignment instructions, etc. can be found on my class web page in Canvas (https://sjsu.instructure.com (https://sjsu.instructucture.com)). You are responsible for regularly checking with the messaging system to learn any updates.

Lecture

Thursday, 6:00 PM to 8:45 PM, ENG 403

Course Description and Requisites

Data representation and preprocessing, proximity, finding nearest neighbors, dimensionality reduction, exploratory analysis, association Object-oriented data organization and representation as strings, arrays, stacks, queues, dequeues, lists, sets, trees, tables, and graphs. Sorting and searching and algorithm design and performance analysis. Testing methods and data are discussed.

Prerequisite(s): Admission into MS Artificial Intelligence, MS Computer Engineering or MS Software Engineering.

CR/NC/I Undergraduate

Classroom Protocols

See class General policies to be posted in the class website.

Major exams in this class may be video recorded to ensure academic integrity. The recordings will only be viewed if there is an issue to be addressed. Under no circumstances will the recordings be publicly released. Recordings are deleted at the end of the semester or earlier as per University regulations.

Your commitment as a student to learning is evidenced by your enrollment at San Jose State University. University Academic Integrity Policy requires you to be honest in all your academic course work. Faculty members are required to report all infractions to the office of Student Conduct and Ethical Development. Check Student Conduct and Ethical Development website for further information.

Course Learning Outcomes (CLOs)

	Description
CLO 1	Discuss and analyze a variety of fundamental concepts and practices in the areas of software engineering.
CLO 2	Apply object-oriented software design methodologies in software programs.
CLO 3	Use standard abstract data types and data structures, including stacks, queues, and linked lists, trees, in the design of software programs.
CLO 4	Apply standard algorithmic techniques including recursions, hashing, searching, and sorting, in the design of software programs.
CLO 5	Use high-level software development tools, including advanced text editors, compilers, linkers, source-level debuggers in implementation and debugging of software programs.

Course Materials

Textbooks

W. Savitch, Problem Solving with C++, 10th edition, Addison-Wesley ISBN-10: 0134448286; ISBN-13: 978-0134448282

D.S. Malik, Data Structures Using C++, 2nd Edition, Course Technology, Cengage Learning, 2010, ISBN-10: 0-324-78201-2, ISBN-13: 978-0-324-78201-1

References

Adam Drozdek, Data Structures and Algorithms in C++, 4th Edition, Cengage Learning, 2012, ISBN-13: 978-1133608424

Michael T. Goodrich, et al, Data Structures and Algorithms in C++, 2nd Edition, Wiley, 2011, ISBN-13: 978-0470383278

Paul Deitel and Harvey Deitel, C++ How to Program, 9th Edition, Prentice Hall, 2013, ISBN-10: 0133378713, ISBN-13: 9780133378719

✔ Grading Information

Grading Policy

• Extra credit options: not available but lowest quiz may be dropped.
• Penalty (if any) for late or missed work: zero grade in missed assignment
• List of the percentage weight assigned to various class assignments:

20% Quizzes (~bi-weekly)
20% Programming Homework (~bi-weekly)
20% Midterm(s)
40% Final

No make-ups exams except in case of verifiable emergency circumstances; once you are back in school, you need to take the exam within a week assuming that you provide documents to justify your absent and it is for a short time. No credit for any late turnings.

Grade bands:

A+ : > 94 A : 90 - 94 A- : 85 - 89
B+ : 80 - 84 B : 75 - 79 B- : 70 - 74
C+ : 65 - 69 C : 60 - 64 C- : 55 - 59
D+ : 50 - 54 D : 45 - 49 D- : 40 - 44
F : < 40 (0.5 - 0.9) = 1 (0.1 - 0.4) = 0

Important: As an example, a student with a final grade of 96 / 100 will have a letter grade of A (not A+). I.e. no special handling of border situations will be performed and the grading scale above will be followed strictly.

"A minimum aggregate GPA of 2.0 SJSU Studies (R, S, & V) shall be required of all students as a graduation requirement." To see full text, review University Policy S11-3 at http://www.sjsu.edu/senate/docs/S11-3.pdf.

CR grade is B- 70 or higher

University Policies

Per University Policy S16-9 (PDF) (http://www.sjsu.edu/senate/docs/S16-9.pdf), relevant university policy concerning all courses, such as student responsibilities, academic integrity, accommodations, dropping and adding, consent for recording of class, etc. and available student services (e.g. learning assistance, counseling, and other resources) are listed on the Syllabus Information (https://www.sjsu.edu/curriculum/courses/syllabus-info.php) web page. Make sure to visit this page to review and be aware of these university policies and resources.

Course Schedule

Week	Topics	Textbook - Savitch	Textbook - Malik
01/22/26	Administrative		
01/29/26	C++ Basics, Flow of Control	1, 2, 3	
	Procedural Abstraction, Functions 1	4	
02/05/26	Functions 2	5	
	I/O Streams	6	
02/12/26	Arrays	7	
	Strings, Vectors	8	
02/19/26	Algorithm complexity, C++ class	10, 17	1
02/26/26	Object-oriented Design: inheritance, polymorphism, overloading.	11, 15	2

Week	Topics	Textbook - Savitch	Textbook - Malik
03/05/26	Pointers, virtual function, array-based list	9	3
	Standard template library.	18	4
03/12/26	*Midterm 1*		
	Linked lists and unordered linked lists.	13	5
	Ordered linked lists and doubly linked lists		5
03/19/26	Problem solving using recursion.	14	6
	Recursion and backtracking	14	6
03/26/26	Stacks		7
	Queues		8
04/09/26	*Midterm 2*		
	Search algorithms		9
	Hashing		9
04/16/26	Simple sorting algorithms		10
	Simple sorting algorithms, Quicksort		10
04/23/26	Merge-sort, Heapsort		10
	Binary search trees and binary tree traversals		11

Week	Topics	Textbook - Savitch	Textbook - Malik
	Binary search trees and binary tree traversals		11
04/29/26	AVL trees		11
	B-Trees		11
05/07/26	Graphs		11
05/14/26	FINAL EXAM (5:30-7:30pm)		

Schedule subject to change with fair notice.